Cyber Physical Robotics

Dr. Paulo Garcia

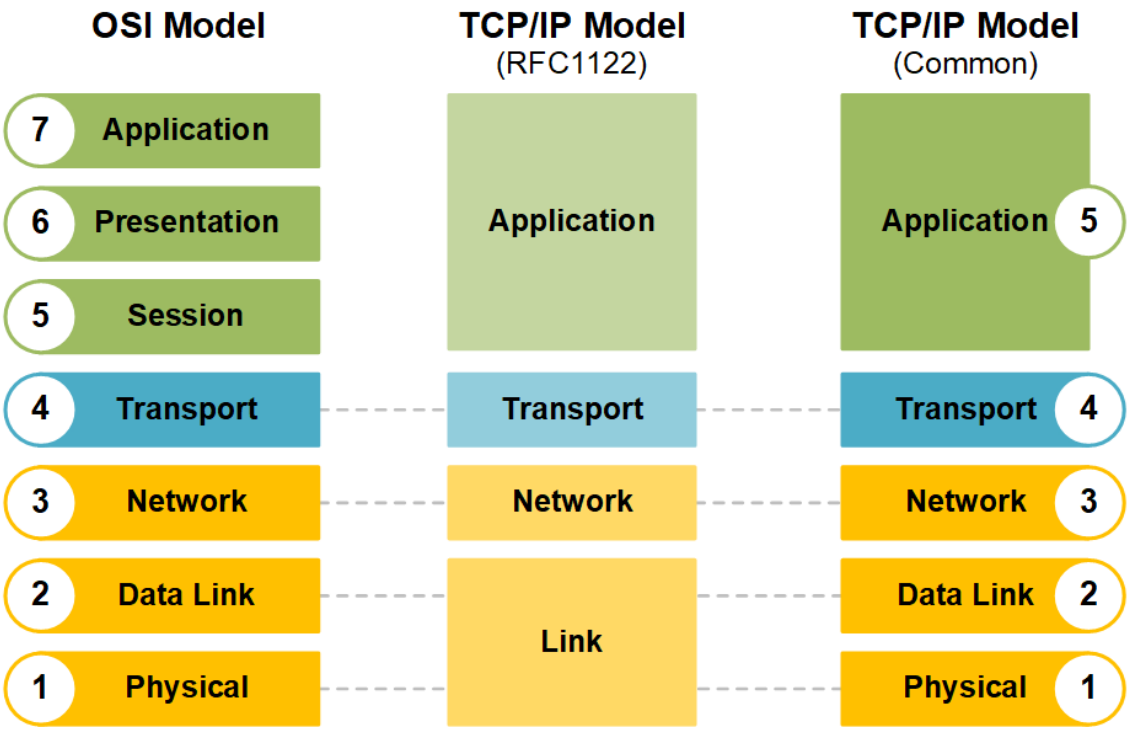
2026

Cyber Physical Robotics

Today:

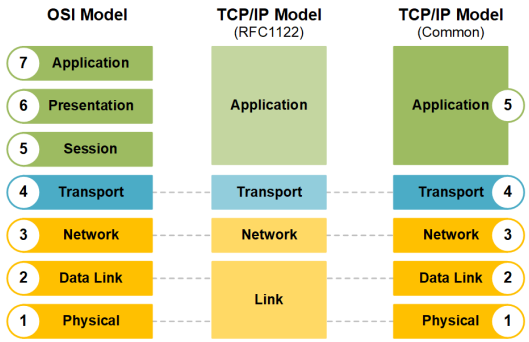
- Network communication
  - The network architecture
  - Sockets as the building block
  - Why communication has variable latency

Cyber Physical Robotics

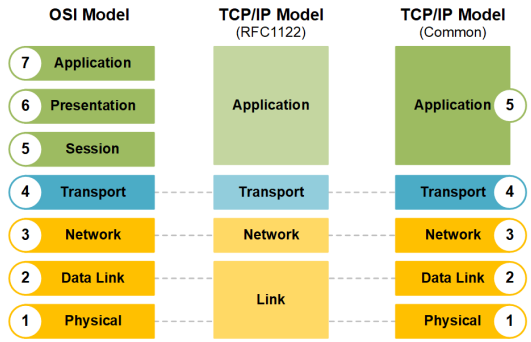


Cyber Physical Robotics

- **Physical layer**
  - Interfacing with the actual medium of communication
  - E.g., writing bits to a wire, or sending wireless data through the air

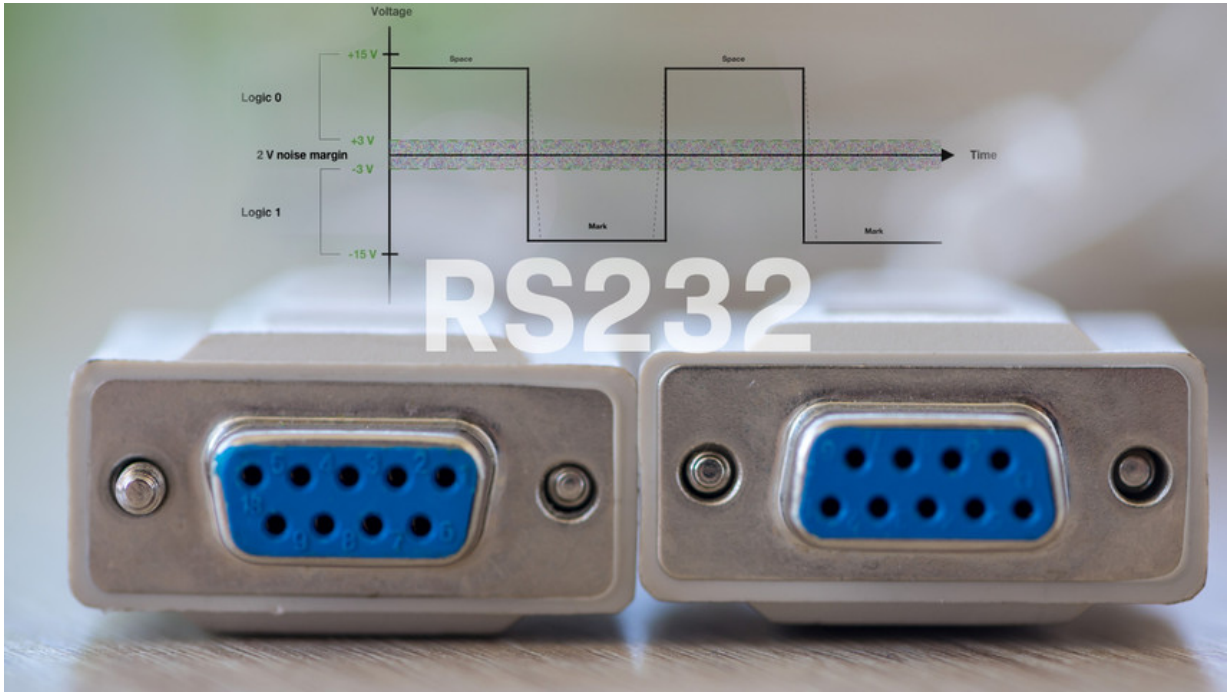


- **Data Link layer**
  - Digital protocol to transfer at the *byte* level
  - E.g., order of bits, types of error correction

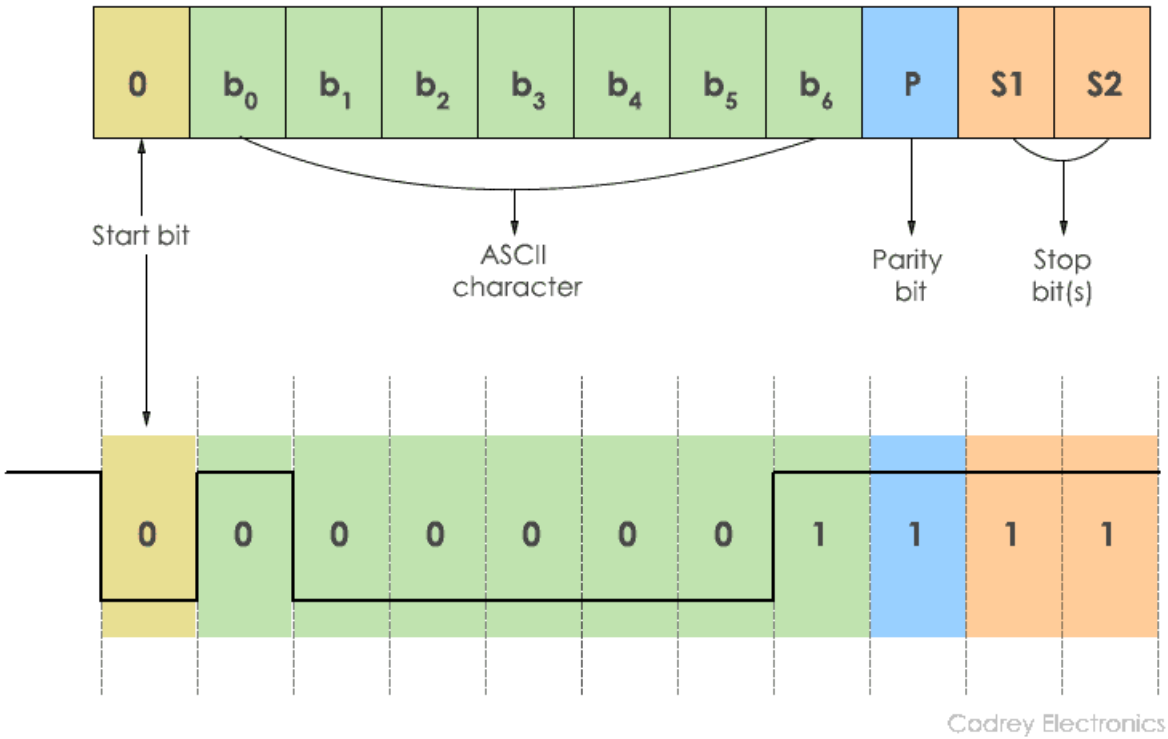


Cyber Physical Robotics

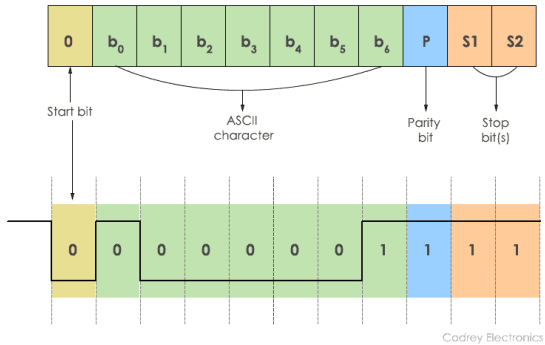
RS-232: the **serial port**



RS-232 protocol: data link layer

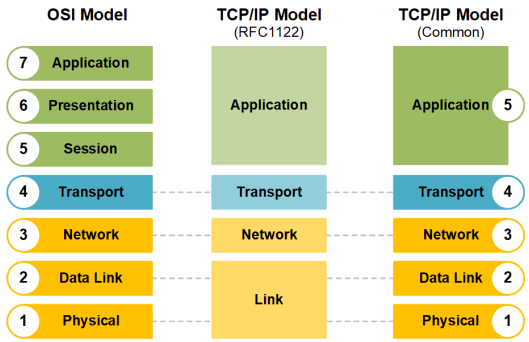


- RS232 protocol
  - Parity bit allows **detecting**, but not correcting, one-bit errors
  - Cannot detect errors of more than 1 bit
  - (see **Hamming codes** for general error-correction)
- Why are there errors?
  - Electromagnetic interference

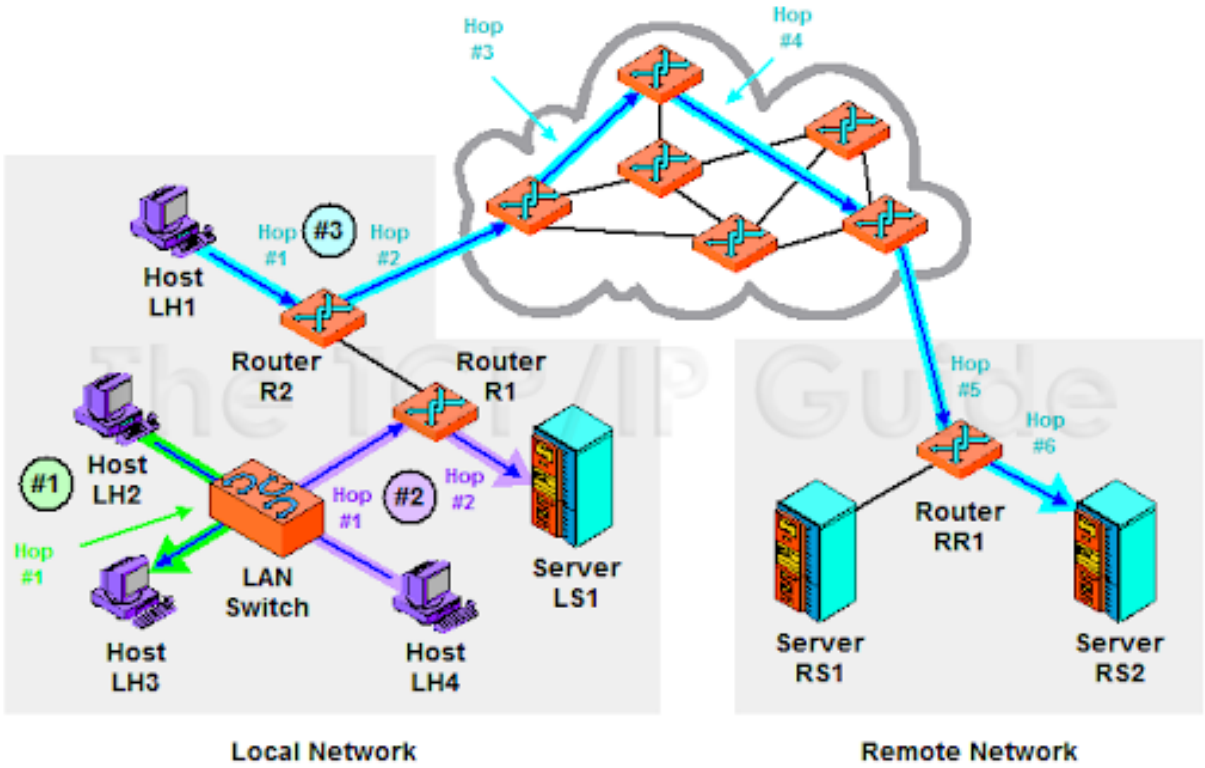


Cyber Physical Robotics

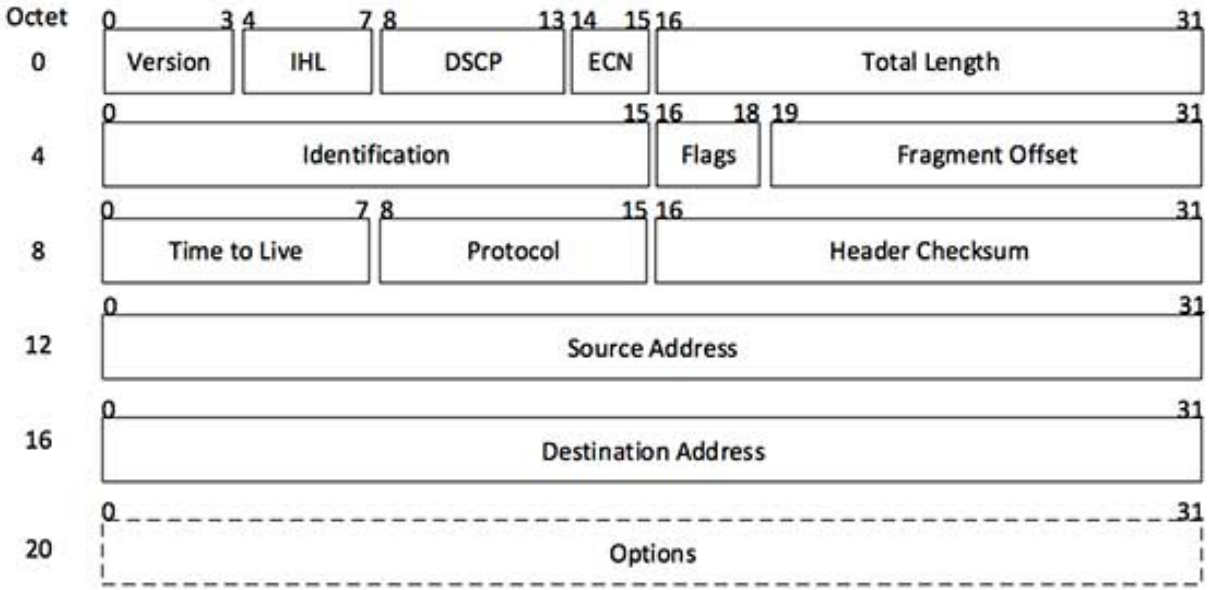
- **Network layer**
  - Tries to transmit messages across multiple nodes
  - (phones, routers, servers, etc.)
  - *assuming there are no errors*



Network layer: IP (Internet Protocol)

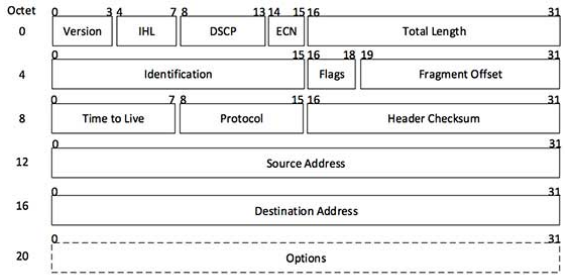


IPv4 (we now use IPv6) message format



[Image: IP Header]

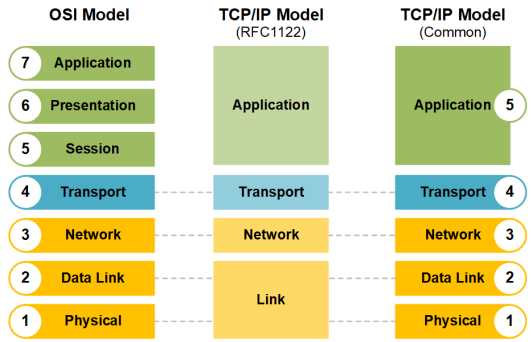
- **Network layer**
  - Every single IPv4 message must be transmitted, **byte by byte**, by the data link layer
  - Hopefully, with error correction
  - At the very least, error detection (messages dropped if wrong)
  - **Does not guarantee delivery, does not guarantee order of messages, does not guarantee messages are not duplicated**



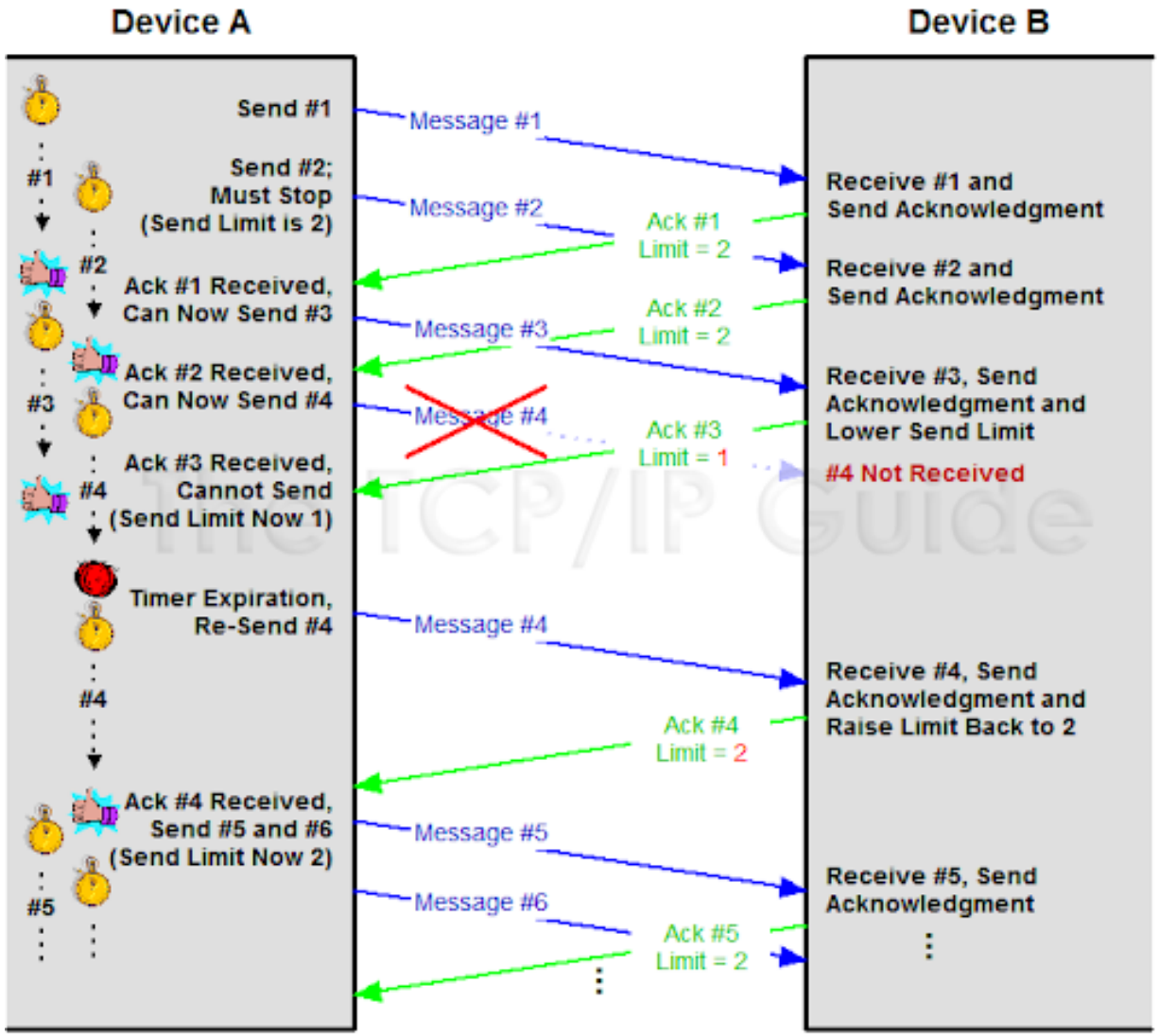
[Image: IP Header]

Cyber Physical Robotics

- **Transport layer**
  - TCP (Transmission Control Protocol)
  - (the entire internet is built on TCP/IP)
  - Guarantees reliable communication, but **does not guarantee timing**

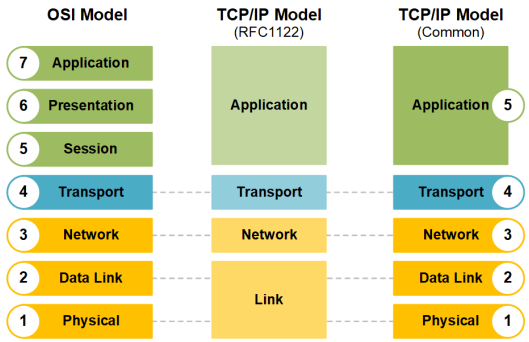


TCP message-passing protocol



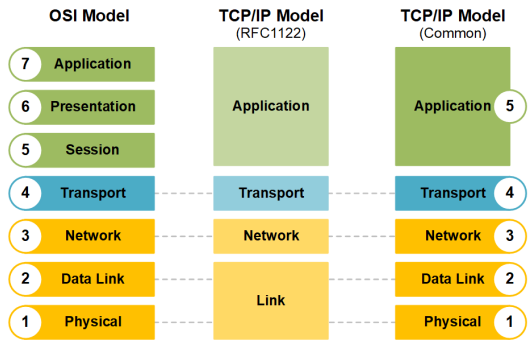
Cyber Physical Robotics

- **Application layer**
  - Internet is built on HTTP
  - Every application is free to do whatever it wants
  - **But everything rests on TCP/IP**



Cyber Physical Robotics

- At the level we work on (IOT, Cyber Physical Robotics)
  - Most of the time, we're working at the TCP/IP level
  - The important thing to remember: we have no way of knowing **when** messages will arrive
  - This is critical for real-time systems



Cyber Physical Robotics

Real-time systems

- Are not about **speed**: real-time does not mean **fast**
- They're about **predictability**
  - Events have an associated **deadline**
  - We need to guarantee all deadlines are met

Cyber Physical Robotics

The problem in Cyber Physical Robots

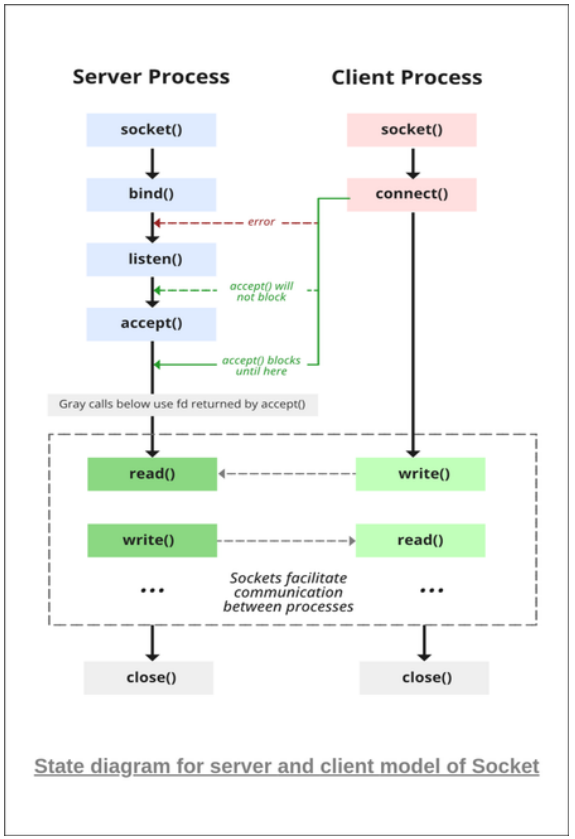
- Robot 1 sees event A (person entered a room): broadcasts that message
- Robot 2 receives message (event A) from robot 1
- Robot 2 sees event B (person leaves the room): broadcasts that message
- Robot 1 receives message (event B) from robot 2
  - **Robots 1 and 2 believe person came in, then went out: not here right now**
- Robot 3 receives message (event B) followed by message (event A)
  - **Robot 3 believes person went out, then came in: is here right now**

TCP/IP (network layer)

- On almost all systems imaginable (Windows, Linux, MacOS...)
- Is built on **sockets**

Cyber Physical Robotics

- Sockets
  - A mechanism provided by libraries/OSs for TCP/IP client-server communication
  - On Linux, provided by sys/socket.h



State diagram for server and client model of Socket

Cyber Physical Robotics

Challenge for you:

- Connect computer 1 to computer 2
- Measure message-passing time

Next week

- Scheduling theory